Analysis of different MOR methods for solving nonlinear Elliptic PDEs

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Abstract

PUNTI DA SCRIVERE (cenni):

- **cosa facciamo e di cosa si tratta**
- **descrizione problema (su cosa facciamo esperimenti numerici): problema ellittico non lineare parametrizzato, condizioni al contorno, configurazioni in analisi**
- **obiettivo: valutare e confrontare POD e PINNs+POD-NN rispetto a full order**
- **risultati: cosa abbiamo ottenuto dalle simulazioni**

I. Introduction

The numerical approximation of nonlinear parametrized partial differential equations (PDEs) using high fidelity techniques, which means solving the full discretized problem, called full-order model (FOM), typically results in very large systems that become computationally expensive and often unfeasible. For this reason, model order reduction (MOR) techniques have gained considerable attention: they are exploited to decrease computational costs while maintaining a good solution approximation.

In the present paper, we study a nonlinear elliptic parametrized equation with homogeneous Dirichlet boundary conditions, on a two-dimensional square domain: both cases with source term dependent on or independent of the parameter are considered. In these two settings, three approaches are implemented[1]:

1. the first is the POD-Galerkin method over a finite element FOM;
2. then we focus on parametric PINN, using the same network structure with either parameters independent and dependent source term;
3. in the end, we use, as another reduce method, the hybrid approach of the POD-NN.

Afterwards, the results in terms of computational costs and accuracy are compared using as a benchmark the FOM results.

The outline of the project is as follows: we first introduce the model definition and the methods used; the numerical studies, results and comparisons are next tackled for the given problems; lastly, the conclusions are discussed.

Before continuing, we assume the reader is proficient enough about functional analysis, to avoid explaining all the Hilbert spaces used in the weak formulation (such as $H_0^1(\Omega)$), and about FEM.

[1]For the acronyms meaning see later sactions.

II. PDE definition and methods

In the following, we talk about the discussed PDE and briefly describe the four approaches studied to solve it.

It's noteworthy that, for the sake of notational brevity, the [explicit] dependency on the parameters $\boldsymbol{\mu}$ in any function f is represented as a superscript: $f^{\boldsymbol{\mu}} \equiv f(\mathbf{x};\boldsymbol{\mu})$.

A. PDE definition

The nonlinear elliptic PDE studied consist of finding

$$u^{\boldsymbol{\mu}}(\mathbf{x}) : \Omega \times \mathcal{P} \to \mathbb{R}$$

such that it's the solution, for some fixed parameters

$$\boldsymbol{\mu} \equiv (\mu_0, \mu_1) \in \mathcal{P} = [0.1, 1]^2$$

in the parametric square domain, of the following problem with homogeneous Dirichlet boundary conditions:

$$[\text{EP}]^{\boldsymbol{\mu}} \begin{cases} -\Delta u^{\boldsymbol{\mu}} + h^{\boldsymbol{\mu}} = g^{\boldsymbol{\mu}} + \dfrac{\mu_0}{\mu_1} & \text{in } \Omega, \\ u^{\boldsymbol{\mu}} = 0 & \text{on } \partial\Omega, \end{cases}$$

where

$$h^{\boldsymbol{\mu}} : V \times \mathcal{P} \to V, \quad (u;\boldsymbol{\mu}) \mapsto h^{\boldsymbol{\mu}}(u) \equiv \frac{\mu_0}{\mu_1} e^{\mu_1 u}$$

is the nonlinear functional term on a space V later defined,

$$g^{\boldsymbol{\mu}}(\mathbf{x}) : \Omega \times \mathcal{P} \to \mathbb{R}$$

is the source term to be defined while

$$\mathbf{x} \equiv (x_0, x_1) \in \Omega = (0,1)^2$$

is the spatial square domain. For the source term two different cases are considered:

C1. independent of $\boldsymbol{\mu}$ with $g^{\boldsymbol{\mu}}(\mathbf{x}) \equiv g_1(\mathbf{x})$ where:

$$g_1(\mathbf{x}) \equiv 100\sin(2\pi x_0)\cos(2\pi x_1) \tag{2}$$

C2. dependent on $\boldsymbol{\mu}$ with $g^{\boldsymbol{\mu}}(\mathbf{x}) \equiv g_2^{\mu_0}(\mathbf{x})$ where:

$$g_2^{\mu_0}(\mathbf{x}) \equiv 100\sin(2\pi\mu_0 x_0)\cos(2\mu_0\pi x_1) \tag{3}$$

Lastly the weak formulation of [EP]$^{\boldsymbol{\mu}}$ is

$$[\mathrm{VP}]^{\boldsymbol{\mu}} \begin{cases} \text{Find } u^{\boldsymbol{\mu}} \in V \text{ such that } \forall v \in V \\ a(u^{\boldsymbol{\mu}}, v) + k^{\boldsymbol{\mu}}(u^{\boldsymbol{\mu}}, v) = F^{\boldsymbol{\mu}}(v), \end{cases}$$

where $V \equiv H_0^1(\Omega)$

$$a : V \times V \to \mathbb{R}, \quad (u,v) \mapsto a(u,v) \equiv \int_\Omega \nabla u \cdot \nabla v;$$

$$k^{\boldsymbol{\mu}} : V \times V \to \mathbb{R}, \quad (u,v) \mapsto k^{\boldsymbol{\mu}}(u,v) \equiv \int_\Omega h^{\boldsymbol{\mu}}(u) v;$$

$$F^{\boldsymbol{\mu}} : V \to \mathbb{R}, \qquad v \mapsto F^{\boldsymbol{\mu}}(v) \equiv \int_\Omega \left(g^{\boldsymbol{\mu}} + \frac{\mu_0}{\mu_1} \right) v,$$

for which we assume that a is continuous and coercive with constants $\gamma \in \mathbb{R} < +\infty$ and $\alpha \in \mathbb{R} > 0$ respectively.

B. Full order model (FOM)

The FOM solution uses the Galerkin method: consider a subspace $V_\delta \subseteq V$ with dimension $N_\delta \equiv \dim\{V_\delta\}_{\delta>0}$, depending on the refinement parameter $\delta > 0$, then [VP]$^{\boldsymbol{\mu}}$ becomes

$$[\mathrm{VP}]_\delta^{\boldsymbol{\mu}} \begin{cases} \text{Find } u_\delta^{\boldsymbol{\mu}}(\boldsymbol{\mu}) \in V_\delta \text{ such that } \forall v_\delta \in V_\delta \\ a(u_\delta^{\boldsymbol{\mu}}, v_\delta) + k^{\boldsymbol{\mu}}(u_\delta^{\boldsymbol{\mu}}, v_\delta) = F^{\boldsymbol{\mu}}(v_\delta), \end{cases}$$

in which now $u_\delta^{\boldsymbol{\mu}}$ approximate the true solution $u^{\boldsymbol{\mu}}$. However, due to the nonlinearity of $k^{\boldsymbol{\mu}}$ it's not possible to obtain a matrix form of [VP]$_\delta^{\boldsymbol{\mu}}$; hence, it's necessary to use an iterative approach, chosen to be the Newton method: assume to know the solution of $u_{\delta,n}^{\boldsymbol{\mu}}$ of the k-th step, the update rule takes the form

$$u_{\delta,n+1}^{\boldsymbol{\mu}} = u_{\delta,n}^{\boldsymbol{\mu}} + \delta u_n^{\boldsymbol{\mu}}. \tag{6}$$

Given the residual of [VP]$_\delta^{\boldsymbol{\mu}}$

$$r^{\boldsymbol{\mu}}(u_\delta^{\boldsymbol{\mu}}, v_\delta) \equiv a(u_\delta^{\boldsymbol{\mu}}, v_\delta) + k^{\boldsymbol{\mu}}(u_\delta^{\boldsymbol{\mu}}, v_\delta) - F^{\boldsymbol{\mu}}(v_\delta) = 0,$$

the direction function δu can be found by having $u_{\delta,n+1}^{\boldsymbol{\mu}}$ such that $r^{\boldsymbol{\mu}}(u_{\delta,n+1}^{\boldsymbol{\mu}}, v) = r^{\boldsymbol{\mu}}(u_{\delta,n}^{\boldsymbol{\mu}} + \delta u_n^{\boldsymbol{\mu}}, v) = 0$, which when linearized gives

$$0 = r^{\boldsymbol{\mu}}(u_{\delta,n}^{\boldsymbol{\mu}} + \delta u, v_\delta) \approx r^{\boldsymbol{\mu}}(u_{\delta,n}^{\boldsymbol{\mu}}, v_\delta) + J_{r,n}^{\boldsymbol{\mu}}[\delta u_n^{\boldsymbol{\mu}}, v_\delta]\delta u_n^{\boldsymbol{\mu}},$$

that is

$$J_{r,n}^{\boldsymbol{\mu}}[\delta u_n^{\boldsymbol{\mu}}, v_\delta]\delta u_n^{\boldsymbol{\mu}} = -r^{\boldsymbol{\mu}}(u_{\delta,n}^{\boldsymbol{\mu}}, v_\delta) \quad \forall v_\delta \in V_\delta, \tag{7}$$

in which $J_{r,n}^{\boldsymbol{\mu}}[\delta u_n^{\boldsymbol{\mu}}, v_\delta]$ is the Jacobian of the residual $r^{\boldsymbol{\mu}}$ evaluated in $u_{\delta,n}^{\boldsymbol{\mu}}$ along the unknown direction $\delta u_n^{\boldsymbol{\mu}}$, namely

$$J_{r,n}^{\boldsymbol{\mu}}[\delta u_n^{\boldsymbol{\mu}}, v_\delta] \equiv \lim_{h\to 0} \frac{r^{\boldsymbol{\mu}}(u_{\delta,n}^{\boldsymbol{\mu}} + h\delta u_n^{\boldsymbol{\mu}}, v_\delta) - r^{\boldsymbol{\mu}}(u_{\delta,n}^{\boldsymbol{\mu}}, v_\delta)}{h}, \tag{8}$$

assuming $r^{\boldsymbol{\mu}}$ is Gateaux differentiable. From (8), it's then clear that

$$J_{a,n}[\delta u_n^{\boldsymbol{\mu}}, v_\delta] = \int_\Omega \nabla \delta u_n^{\boldsymbol{\mu}} \nabla v_\delta,$$

for the linearity of $a(u_\delta^{\boldsymbol{\mu}}, v_\delta)$ w.r.t. $u_\delta^{\boldsymbol{\mu}}$, whereas

$$J_{F,n}^{\boldsymbol{\mu}}[\delta u_n^{\boldsymbol{\mu}}; v_\delta] = 0$$

as $F^{\boldsymbol{\mu}}$ it does not depend on $u_\delta^{\boldsymbol{\mu}}$; thus, the only Jacobian necessary to calculate explicitly is

$$\begin{aligned} J_{k,n}^{\boldsymbol{\mu}}[\delta u_n^{\boldsymbol{\mu}}, v_\delta] &= \lim_{h\to 0} \frac{1}{h} \left[\int_\Omega \frac{\mu_0}{\mu_1} \left(e^{\mu_1 h \delta u} - 1 \right) e^{\mu_1 u_{\delta,n}^{\boldsymbol{\mu}}} v_\delta \right] \\ &= \lim_{h\to 0} \frac{1}{h} \left[\int_\Omega \frac{\mu_0}{\mu_1} (1 + \mu_1 h \delta u - 1) e^{\mu_1 u_{\delta,n}^{\boldsymbol{\mu}}} v_\delta \right] \\ &= \int_\Omega \mu_0 \delta u e^{\mu_1 u_{\delta,n}^{\boldsymbol{\mu}}} v_\delta, \end{aligned}$$

where $e^{\mu_1 h \delta u} \approx 1 + \mu_1 h \delta u$ is valid only with $h \ll 1$ as is the case when $h \to 0$. Therefore, (7) becomes

$$[\mathrm{NP}]_\delta^{\boldsymbol{\mu}} \begin{cases} \text{Find } \delta u_n^{\boldsymbol{\mu}} \in V_\delta \text{ such that } \forall v_\delta \in V_\delta \\ a(\delta u_n^{\boldsymbol{\mu}}, v_\delta) + \mu_0 b(\delta u_n^{\boldsymbol{\mu}}, v_\delta e^{\mu_1 u_{\delta,n}^{\boldsymbol{\mu}}}) = \\ \quad -a(u_{\delta,n}^{\boldsymbol{\mu}}, v_\delta) - \dfrac{\mu_0}{\mu_1} b(e^{\mu_1 u_{\delta,n}^{\boldsymbol{\mu}}}, v_\delta) + F^{\boldsymbol{\mu}}(v_\delta), \end{cases}$$

in which

$$b : V \times V \to \mathbb{R}, \quad (u,v) \mapsto b(u,v) \equiv \int_\Omega uv.$$

Introducing now a set of basis function $\{\varphi_k\}_{k=1}^{N_\delta}$ such that $V_\delta = \mathrm{span}\{\varphi_k\}_{k=1}^{N_\delta}$, leads to the linear system

$$\mathbf{S}_n^{\boldsymbol{\mu}} \delta \mathbf{u}_n^{\boldsymbol{\mu}} = \mathbf{y}_n^{\boldsymbol{\mu}}, \tag{10}$$

where

$$\mathbf{S}_n^{\boldsymbol{\mu}} \equiv \mathbf{S}^{\boldsymbol{\mu}}(\mathbf{u}_{\delta,n}^{\boldsymbol{\mu}}) \in \mathbb{R}^{N_\delta \times N_\delta} \quad \text{and} \quad \mathbf{y}_n^{\boldsymbol{\mu}} \equiv \mathbf{y}^{\boldsymbol{\mu}}(\mathbf{u}_{\delta,n}^{\boldsymbol{\mu}}) \in \mathbb{R}^{N_\delta}$$

have the following definition

$$\begin{aligned} \mathbf{S}_n^{\boldsymbol{\mu}} &\equiv \mathbf{A} + \mu_0 \mathbf{B}_n^{\mu_1}, \\ \mathbf{y}_n^{\boldsymbol{\mu}} &\equiv -\mathbf{a}_n - \frac{\mu_0}{\mu_1}(\mathbf{b}_n^{\mu_1} - \mathbf{b}) + \mathbf{g}^{\boldsymbol{\mu}}. \end{aligned}$$

Some remarks ensue from (10):

- ◇ The update rule in (6) becomes

$$\mathbf{u}_{\delta,n+1}^{\boldsymbol{\mu}} = \mathbf{u}_{\delta,n}^{\boldsymbol{\mu}} + \delta \mathbf{u}_n^{\boldsymbol{\mu}} \tag{11}$$

  where now every term is are vectors carrying the weights associated to the basis functions;
- ◇ $\mathbf{A}$ is the stiffness matrix ($A_{ij} \equiv a(\varphi_j, \varphi_i)$), $\mathbf{b}$ is $\mathbf{b}_j \equiv b(1, \varphi_j)$ while $\mathbf{b}_n^{\mu_1}$ is $(\mathbf{b}_n^{\mu_1})_j \equiv b(e^{\mu_1 u_{\delta,n}^{\boldsymbol{\mu}}}, \varphi_j)$;
- ◇ $\mathbf{A}$, $\mathbf{b}$ and $\mathbf{g}$ are always affine w.r.t. $\boldsymbol{\mu}$;
- ◇ $\mathbf{B}_n^{\mu_1}$ and $\mathbf{b}_n^{\mu_1}$ are always *not* affine w.r.t. μ_1;
- ◇ $\mathbf{g}_n^{\boldsymbol{\mu}}$ is affine w.r.t. μ_0 if $g \equiv g_1$ but *not* if $g \equiv g_2$.

C. Proper Orthogonal Decomposition (POD)

POD is strategy that aims to reduce computational costs by solving the system (10) in a reduced space $V_\rho \subseteq V_\delta$ of dimension $N_\rho \ll N_\delta$: assume to know the basis function $\{\xi_k\}_{k=1}^{N_\rho} \in V_\delta$ of such reduced space (i.e. $V_\rho = \text{span}\{\xi_k\}_{k=1}^{N_\rho}$); then it's possible to express each function w.r.t. $\{\varphi_k\}_{k=1}^{N_\delta}$, obtaining the basis matrix

$$\mathbb{B} = [\boldsymbol{\xi}^1 \cdots \boldsymbol{\xi}^{N_\rho}] \in \mathbb{R}^{N_\delta \times N_\rho};$$

given a vector $v \in \mathbb{R}^{N_\rho}$, one has $\mathbb{B}v \in \mathbb{R}^{N_\delta}$, that is $\mathbb{B}$ can be seen as a projection from V_ρ to V_δ and the same can be said for the inverse projection with $\mathbb{B}^\top$; thus, defining $\delta\hat{\mathbf{u}}_n^{\boldsymbol{\mu}}$ such that $\delta\mathbf{u}_n^{\boldsymbol{\mu}} = \mathbb{B}\delta\hat{\mathbf{u}}_n^{\boldsymbol{\mu}}$, (10) becomes:

$$\hat{\mathbf{S}}_n^{\boldsymbol{\mu}} \delta\hat{\mathbf{u}}_n^{\boldsymbol{\mu}} = \hat{\mathbf{y}}_n^{\boldsymbol{\mu}}, \tag{12}$$

in which $\hat{\mathbf{S}}_n^{\boldsymbol{\mu}} \equiv \mathbb{B}^\top \mathbf{S}_n^{\boldsymbol{\mu}} \mathbb{B}$ and $\hat{\mathbf{y}}_n^{\boldsymbol{\mu}} \equiv \mathbb{B}^\top \mathbf{y}_n^{\boldsymbol{\mu}}$. The strategy can be summarized using the following commutative diagram:

$$\begin{tikzcd}
V_\delta \arrow[r, "\mathbf{S}_n^{\boldsymbol{\mu}}\delta\mathbf{u}_n^{\boldsymbol{\mu}}=\mathbf{y}_n^{\boldsymbol{\mu}}"] \arrow[d, "\mathbb{B}^\top"'] & V_\delta \\
V_\rho \arrow[r, "\hat{\mathbf{S}}_n^{\boldsymbol{\mu}}\delta\hat{\mathbf{u}}_n^{\boldsymbol{\mu}}=\hat{\mathbf{y}}_n^{\boldsymbol{\mu}}"] & V_\rho \arrow[u, "\mathbb{B}"']
\end{tikzcd},$$

where one hopes the reduced path is less computationally demanding than the full one. Of course, as mentioned before, both the iterative and non-affine nature of (10) hinders this chance, though only numerical studies in § III can confirm if that's truly the case.

To find $\mathbb{B}$ consider $M \ll N_\delta$, but large enough so that $M \gg N_\rho$, FOM solutions $\mathbf{u}_\delta^n \equiv \mathbf{u}_\delta^{\boldsymbol{\mu}_n}$, $1 \le n \le M$, of $[\text{VP}]_\delta^{\boldsymbol{\mu}}$ with uniformly varying parameters $\boldsymbol{\mu}$, called snapshots, and define

$$\mathbb{U} \equiv \begin{bmatrix} \mathbf{u}_\delta^1 & \cdots & \mathbf{u}_\delta^M \end{bmatrix}^\top \in \mathbb{R}^{M \times N_\delta}$$

as the matrix whose rows are the snapshots.

Much of the information inside $\mathbb{U}$ is redundant, so one then needs to compress it via the correlation matrix

$$\mathbf{C} \equiv \mathbb{U}\mathbf{A}\mathbb{U}^\top \in \mathbb{R}^{M \times M}; \tag{13}$$

indeed, by solving the eigenvalue problem

$$\mathbf{C}\boldsymbol{\omega}^i = \lambda_i \boldsymbol{\omega}^i, \quad 1 \le i \le M, \tag{14}$$

it holds

$$\text{err}_{M,\rho}^2 = \sum_{i=1}^{M} \|u_\delta^i - P_\rho(u_\delta^i)\|_V^2 = \sum_{i=N_\rho+1}^{M} \lambda_i, \tag{15}$$

where $P_\rho : V_\delta \to V_\rho$ is the projection operator to the reduced space. In essence (15) says that the most important eigenvectors, *i.e.* those that retain all significant variance, are those with the largest eigenvalues; and since $\mathbf{C}$ is real, symmetric and positive semi-definite, we can order the all-positive eigenvalues as

$$\lambda_1 \ge \cdots \ge \lambda_M \ge 0$$

and take only the first N_ρ so that $\text{err}_{M,\rho}^2 \approx 0$.

However, the eigenvectors $\{\boldsymbol{\omega}^k\}_{k=1}^{N_\rho}$ are still in $\mathbb{R}^M$ while $\{\boldsymbol{\xi}^k\}_{k=1}^{N_\rho}$ lie in $\mathbb{R}^{N_\delta}$; to transform them accordingly it is sufficient to notice how (16) changes with (13):

$$\begin{aligned}
(16) \implies & \mathbb{U}\mathbf{A}\mathbb{U}^\top \boldsymbol{\omega}^i = \lambda_i \boldsymbol{\omega}^i, \\
\implies & \mathbf{A}\mathbb{U}^\top \boldsymbol{\omega}^i = \lambda_i \mathbb{U}^\top \boldsymbol{\omega}^i, \\
\implies & \mathbf{A}\boldsymbol{\chi}^i = \lambda_i \boldsymbol{\chi}^i, \quad 1 \le i \le M,
\end{aligned} \tag{16}$$

where $\boldsymbol{\chi}^i = \mathbb{U}^\top \boldsymbol{\omega}^i$, $1 \le i \le N_\rho$, is the transformation sought which also proves $\boldsymbol{\chi}^i$ are the eigenvectors of the stiffness matrix [2].

The last step comes from linear algebra: in fact, we know thnaks to the spectral theorem that both $\boldsymbol{\chi}^k{}_{k=1}^{N_\rho}$ and $\{\boldsymbol{\omega}^k\}_{k=1}^{N_\rho}$ are only orthogonal to one another [1, §6.3 p. 286]; hence the reduced basis vectors are obtained by simply normalizing them:

$$\boldsymbol{\xi}^i \equiv \frac{\boldsymbol{\chi}^i}{\|\boldsymbol{\chi}^i\|_V}.$$

D. Physics-Informed Neural Networks (PINN)

In nonlinear frameworks, to overcome some of the limitations of the POD method and to exploit the advantages of the division in online and offline phases, the recent advances in machine learning have opened new opportunities for solving PDEs. In particular, PINN is a flexible approach that directly uses the equations into the training process of a neural network, it incorporates physical laws given by the differential equations into the loss functions to take advantages in the learning process. This allows PINNs to deal with complex and parametric problems in a data efficient and mesh free way.

Having a general parametric nonlinear partial differential equation, PINNs are neural networks that are trained to solve supervised learning tasks while respecting the laws of physics described by the given equations. The main idea is to incorporate into the loss function both the information obtained from the data and the physical model. The physics-informed term is like a prior knowledge that provides more information to the PINN without the need for more measurements, but only by evaluating the residual of the PDE at additional points in the domain. In our case, we want to approximate the solution of the PDE by considering two contributions to the loss function: one related to boundary information and the other to residual information.

The general structure of the parametric PINN method can be summarized as follows:

$$\begin{aligned}
\mathcal{F}_{NN}: \quad & \Omega \times \mathcal{P} \to \mathbb{V}^{\boldsymbol{\mu}}, \\
& (x, \boldsymbol{\mu}) \mapsto \hat{u}(x, \boldsymbol{\mu}) \approx u(x, \boldsymbol{\mu}).
\end{aligned} \tag{17}$$

[2]The stiffness matrix shares many properties of $\mathbf{C}$: by definition ($A_{ij} \equiv a(\varphi_j, \varphi_i)$) it's real, symmetric and due to the coercivity of a ($a(v,v) \ge \alpha\|v\|_V^2 \ \ \forall v \in V$) it's also definite positive.

where $u(x,\boldsymbol{\mu})$ is the exact solution of the PDE and $\mathbb{V}^{\boldsymbol{\mu}}$ is the functional space where the solution lies. PINNs use optimization algorithms to update the network parameters until the value of the loss function is decreased to a prespecified level.

Given the neural network $v(x,\boldsymbol{\mu}) \in \mathcal{F}_{NN}$, the loss function is defined as:

$$\mathrm{MSE} = \mathrm{MSE}_b^{\boldsymbol{\mu}} + \lambda \mathrm{MSE}_p^{\boldsymbol{\mu}} \tag{18}$$

where $\mathrm{MSE}_b^{\boldsymbol{\mu}}$ is the boundary loss, $\mathrm{MSE}_p^{\boldsymbol{\mu}}$ is the physical loss and λ is the hyperparameter that balances the tradeoff between the two contributions, to be set before the network training.

In particular, if the physical system we are studying is described by a PDE of the form $\mathcal{R}(u(x,\boldsymbol{\mu})) = 0$, the errors are given by:

$$\mathrm{MSE}_b^{\boldsymbol{\mu}} = \frac{1}{N_b}\sum_{k=1}^{N_b} |v(x_k^b,\boldsymbol{\mu}_k^b) - u(x_k^b,\boldsymbol{\mu}_k^b)|^2,$$

$$\mathrm{MSE}_p^{\boldsymbol{\mu}} = \frac{1}{N_p}\sum_{k=1}^{N_p} |\mathcal{R}(v(x_k^p,\boldsymbol{\mu}_k^p))|^2$$

for $(x_k^b,\boldsymbol{\mu}_k^b) \in (\partial\Omega,\mathcal{P})$ and $(x_k^p,\boldsymbol{\mu}_k^p) \in (\Omega,\mathcal{P})$.

Finally, the solution approximation is given by

$$\hat{u}(x,\boldsymbol{\mu}) := \underset{v\in\mathcal{F}_{NN}}{\operatorname{argmin}} \mathrm{MSE}. \tag{19}$$

E. POD-NN hybrid approach

By combining the previous two methods, we obtain a promising paradigm: the POD-NN hybrid approach. This leverages the dimensionality reduction given by POD to obtain a compact representation of the solution space while using a neural network to learn the parametric dependence of the reduced coefficients. This strategy maintains the accuracy of POD while exploiting the benefits of neural networks, providing a computationally efficient alternative to both the MOR and PINN approaches.

POD-NN is a strategy that provides a fast tool to deal with non affine structures. It avoids the online projection by exploiting the direct projection of snapshots onto the reduced space built with the POD method, and then using them to create and train a neural network.

The general structure of the network can be summarized as follows:

$$\begin{aligned} \pi_{NN}: \quad & \mathcal{P} \to \mathbb{R}^N, \\ & \boldsymbol{\mu}^* \mapsto \boldsymbol{u}_N^{NN}(\boldsymbol{\mu}^*) \end{aligned} \tag{20}$$

where N is the dimension of the reduced space obtained by applying the POD. The inputs are the parameters of the training set, while the output is the Galerkin projection of the snapshots. For the output, which has the form $\boldsymbol{u}_N^{NN}$, the following relation holds:

$$\mathbb{B}\boldsymbol{u}_N^{NN}(\boldsymbol{\mu}) \approx \mathbb{P}^{\boldsymbol{\mu}}\boldsymbol{u}_\delta(\boldsymbol{\mu})$$

where $\mathbb{B} \in \mathbb{R}^{N_\delta \times N}$ is the basis functions matrix of the reduced space, and $\mathbb{P}^{\boldsymbol{\mu}} = \mathbb{B}\mathbb{P}_{\delta\to N} \in \mathbb{R}^{N_\delta\times N_\delta}$ is composed by the projection operator, which leads to the best approximation of $u_\delta^{\boldsymbol{\mu}}$ in the reduced space, pre-multiplied by the basis function matrix. This allows us to compute $\boldsymbol{u}_N^{NN}(\boldsymbol{\mu})$ for each snapshot without solving the reduced system, obtaining the best approximation to $\boldsymbol{u}_\delta$ because we have a direct connection between $\boldsymbol{u}_\delta^{\boldsymbol{\mu}}$ and $\boldsymbol{u}_N^{NN}(\boldsymbol{\mu})$. To train the network, we minimize the loss given by the distance between the exact u_N solution and the network approximation:

$$\mathrm{MSE} = \frac{1}{N_{snap}} \sum_{\boldsymbol{\mu}\in\mathcal{P}_{train}} ||\boldsymbol{u}_N^{NN}(\boldsymbol{\mu}) - \boldsymbol{u}_N(\boldsymbol{\mu})||^2, \tag{21}$$

where N_{snap} is the number of data in the training set.

To sum up, we can compute offline the snapshots collection, apply the POD strategy and train the network. Then, we evaluate the network during the online phase. In this way, we exploit the POD properties and the online phase becomes very fast. In contrast, the offline phase is longer and the accuracy is more difficult to control.

III. Numerical studies

In this section the reduced solution seen in § III are compared in terms of computational costs and accuracy with respect to the FOM one.

Introduzione con

- ◇ **'set up' di simulazione**
- ◇ **metriche usate per valutare risultati**
- ◇ **confronti tra gli esperimenti**

Esperimenti nello specifico: confronti risultati con grafici

A. Source term independent of parameters

B. Source term dependent on parameters

IV. Conclusions

References

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